

**DEPARTMENT OF ENVIRONMENTAL QUALITY
OIL, GAS, AND MINERALS DIVISION**

**RESPONSE TO PUBLIC COMMENTS
WEXFORD WATER TECHNOLOGIES, LLC
MINERAL WELL APPLICATION
NON-HAZARDOUS WASTE DISPOSAL WELL
CEDAR CREEK TOWNSHIP, WEXFORD COUNTY, MICHIGAN**

INTRODUCTION

Wexford Water Technologies, LLC (WWT) has submitted a permit application to the Michigan Department of Environmental Quality (DEQ) to drill and operate a disposal well (WWT1) for the disposal of non-hazardous landfill leachate generated on-site at Wexford County Landfill. Under Michigan law, the permitting, construction, drilling, and operation of this well is subject to the provisions of Michigan's Mineral Well program, Part 625 of the Natural Resources and Environmental Protection Act, 1994 PA 451, as amended (NREPA), and the administrative rules established under Part 625.

Disposal wells of this type are permitted by the DEQ Oil, Gas, and Minerals Division (OGMD) and the U.S. Environmental Protection Agency (EPA), Underground Injection Control (UIC) Program. According to the EPA classification, this proposed well is considered a Class I disposal well. Class I disposal wells inject industrial and municipal wastewaters into deep, confined formations far below and isolated from underground sources of drinking water (USDW).

Wexford County Landfill is located just north of Cadillac, Michigan, and south of Manton. WWT proposes to dispose non-hazardous leachate from the landfill into deep, confined, porous rock formations between 3,100 and 6,000 feet below the ground surface. The disposal rock formations naturally contain high levels of dissolved minerals. The base of the lowest known freshwater aquifer is approximately 900 feet below the ground surface. The proposed WWT 1 disposal well is designed to isolate the disposal zone from the freshwater zones with multiple strings of casing (steel pipes) cemented to the surface. The disposal is through steel tubing within the casing strings. The disposal zone and the freshwater zone are separated vertically by more than 2,000 feet, with several confining rock formations in between. Under no circumstances would a disposal well be approved or allowed to operate if there were indications that the disposed fluids could migrate into freshwater zones.

PUBLIC ENGAGEMENT

Due to the anticipated public interest regarding this proposed activity, OGMD required WWT to place notice of the project in two newspapers with a general circulation in the area. Notice was published in the *Traverse City Record Eagle* on October 27, 2017 and the *Northern Michigan News* on October 30, 2017. The OGMD posted the permit application and details regarding the project on its website. The November 22, 2017 deadline for public comment was posted on the DEQ calendar on November 2, 2017. This deadline was concurrent with EPA's public notice deadline. Comments

received included requests for a public hearing. Subsequently, a decision was made to hold one public meeting and hearing with staff from both OGMD and EPA. This meeting was held on March 19, 2018, at the Cadillac Senior High School Auditorium and was attended by an estimated 150 people. The purpose of this forum was to invite and record public comment, inform the area residents of the project, and answer questions regarding the proposed activities. Notice of the public meeting and hearing were posted on the DEQ Calendar, on the OGMD website, and in newspapers of local circulation. Notice was published in the *Traverse City Record Eagle* on February 25, 2018, *Cadillac News* March 3, 2018, and *Northern Michigan News* on March 5, 2018. At the same time that notice of the public meeting and hearing was made, OGMD and EPA held a second public comment period. OGMD accepted comments through March 28, 2018. Written comments could be submitted via an OGMD email inbox or through the U.S. Mail. Some comments were received after the comment period expired but were still reviewed and considered in the final decision. In total, the OGMD received over 200 comments from 50 commenters related to this project.

RESPONSE TO COMMENTS

In this document, the OGMD is not responding directly to individual comments either in support of or in opposition to the proposal. The supportive comments were generally centered around the benefits of promoting economic growth and in support of developing one's property. Comments received were predominantly against this proposed well. Many commenters expressed their wish that a permit be denied. For this response document, opposing comments are grouped into the following categories: characterization of the waste, current contamination from the landfill, environmental risk, operator liability, injection well construction, testing, and monitoring, and alternative treatments.

WASTE CHARACTERIZATION

WWT proposes to use this well for the disposal of non-hazardous landfill leachate wastewater. Landfill leachate is the liquid generated within a landfill. It is created from moisture contained within the waste itself, and the precipitation that falls on the landfill, which moves through the waste. Modern landfills are constructed with liners to collect leachate at the bottom of the landfill. Leachate is primarily water, with low concentrations of dissolved compounds that do not cause the leachate to be classified as hazardous under Michigan law. Storm water generated on-site and fluids derived from the operation and maintenance of the well also make up part of this wastewater. The pH of the wastewater is generally neutral, and is not considered ignitable, reactive, or corrosive. The wastewater will be tested regularly to monitor for various parameters and assure that it remains classified as non-hazardous. WWT has provided a Waste Analysis Plan (WAP) which it will follow to meet the testing requirements. The sampling frequency is primarily quarterly, although there are a few parameters that will be sampled and tested annually. A copy of the WAP has been available on the OGMD website.

WWT submitted waste analysis with its original permit application for the WWT 1 well in August of 2015. Commenters expressed concern that the waste analysis from 2015 was outdated and did not include ammonia as an analyte. OGMD required WWT to submit updated waste analysis results, and although not required, the new analytical included ammonia. Upon review of the new data, OGMD found that the sampling results were consistent with the 2015 analytical data.

CURRENT CONTAMINATION

Due to groundwater contamination from past practices at the landfill, many residents living close to the landfill are concerned about the environmental risk the proposed waste disposal well poses. The historic groundwater contamination plumes are isolated within the uppermost sand aquifer, with a confining clay layer separating the lower and upper sand aquifer. This clay layer ranges in thickness from 3 feet to 30 feet thick.

After initial review of the WWT 1 well application, OGMD had concerns over the proposed disposal well location in the northern half of the landfill. It was located within an area of groundwater contamination, posing a risk of carrying the shallow contamination into a deeper, currently unused USDW during drilling. OGMD required WWT move the location outside of the area of known contamination. WWT withdrew its original application in August of 2016 and applied for a new location to the south - the current location of the proposed WWT 1 well. The applicant submitted supplemental information in the form of a Technical Memorandum showing the proposed location is not within an area of existing groundwater contamination. The proposed well will not be drilled through the known shallow contamination plume.

Part 625 Mineral Well rules require surface casing to be set 100 feet below all known freshwater aquifers and into competent bedrock. To further ensure additional protection to freshwater resources, OGMD recommended WWT set two casing strings (steel pipes) through the freshwater zones and cementing each over the entire length – back to the surface. The first casing string will be set into a confining clay layer located at a depth of approximately 370 feet and the second casing string will be set at approximately 1600 feet into a shale layer. These two initial casing strings will prevent groundwater flow between the upper and lower aquifers.

ENVIRONMENTAL RISK

Concerns were expressed regarding the potential for contamination of water resources. To protect water resources from contamination, Michigan's mineral well program has strict and robust casing and sealing requirements. As detailed in the section above, OGMD is requiring WWT go beyond state rules by setting two strings of casing through freshwater zones. Overall this well will be constructed with multiple concentric casing strings (steel pipes) with the uppermost three strings cemented over the entire length - back to the surface. By cementing the three casing strings to surface, the freshwater zones are protected from fluid migration upward from the disposal zone.

During well drilling operations, above ground steel tanks will be used for the handling of drilling mud and cuttings. To further assure that no releases to the environment occur, WWT is required to have secondary containment placed around the well(s) and any future on-site tanks or surface facilities.

After the well is drilled, state and federal rules require mechanical integrity be demonstrated prior to being approved for injection of fluid. Mechanical integrity tests ensure that there are no significant leaks

or unwanted fluid movement associated with the well. Periodic mechanical integrity testing is required for the life of the well. Also, WWT is required to continuously monitor injection pressure, pressure in the space between concentric casings or between the outermost casing and the borehole wall (termed the “annulus”), flow rate, and cumulative injected volume, and report those figures to OGMD monthly.

Finally, in the unlikely event of a spill or release, Michigan operators are required to promptly report the details of that spill or release and begin recovery and cleanup efforts immediately. Since Michigan’s mineral well statute was enacted in 1969, there have been no instances of a subsurface leak from a wellbore impacting freshwater. There have been some instances of spills from pipelines or storage facilities associated with disposal wells, but most have been minor, discovered and reported quickly, and addressed in a timely manner.

OPERATOR LIABILITY

Many commenters expressed concern that Wexford Water Technologies was lacking in overall liability and financial assurance. They believe the conformance bond amount is inadequate. WWT provided conformance bonding in the amount of \$33,000 for this well. This bonding is in place to cover the costs for plugging the well and restoring the site location, should that need arise. State regulations do not require that any additional bonding be imposed beyond the amounts defined in the rules. WWT has met those bonding requirements. Beyond conformance bonding, most prudent operators also carry liability insurance to cover unforeseen events. Although it is beyond our purview, it is likely that this would be the case should this project move forward.

There has been public interest concerning the ownership of Wexford Water Technologies LLC and its structure as a Limited Liability Company (LLC). As required, WWT has submitted an organization report identifying itself as a subsidiary of Wexford County Landfill, LLC, and principals Edward Ascione and Michael Ascione as co-owners. Part 625 rules do not address company structure. The LLC company structure is commonly used by permittees of wells permitted by OGMD. Being an LLC does not mean that WWT can avoid responsibility for environmental liabilities. LLC organizations have been and will be held liable for violations of regulations or statutes. Moreover, WWT would be required to clean up any contamination due to the activities associated with this project, pursuant to Part 625 Mineral Wells and Part 201 Environmental Remediation of the NREPA.

INJECTION WELL CONSTRUCTION, TESTING, AND MONITORING

Injection wells are designed and constructed to allow disposal while preventing the movement of disposed wastewaters into shallower freshwater zones. Wells that meet design and operating requirements are very low risk. There are redundant protections and strict requirements related to well location, design, construction, operation, monitoring, testing, reporting, and closure. The safeguards in place ensure protection to freshwater, the environment, and human health.

Injection wells must be constructed with multiple concentric casing strings (steel pipes). Once the hole is drilled to the depth that a certain casing will be set, the casing is placed in the hole. The annulus is then cemented to secure the casing into place and to seal off the drilled formations. This cement is initially pumped through the inside of the casing. As pumping continues, cement is circulated out and around the outside of the casing (through the annulus), up to the surface. To assure sufficient cementing amounts, the annulus volume is calculated and a certain volume in excess of that volume is required by permit. Once enough cement reaches the surface from the annulus of the well, there is a required wait to allow time for the cement to harden and set. The WWT 1 well will have three concentric steel casings. Each interior casing will be set deeper than the previous and each annulus will be cemented to surface. Removable steel tubing is placed inside the innermost cemented casing. The tubing is held in place and sealed down-hole by a tool called a packer.

Prior to injecting leachate into the disposal well, WWT will be required to demonstrate the internal and external mechanical integrity of the well. Internal mechanical integrity means the well has no significant leak in the casing, tubing, packer, or wellhead. A pressure test can be performed to determine whether there are any leaks. External mechanical integrity means the well has no significant fluid movement through vertical channels adjacent to the wellbore. External mechanical integrity is determined using wireline logs, where measurements devices are lowered into the well on a cable. Most commonly, a log measuring temperature is used to detect abnormal conditions potentially caused by a leak or fluid movement. WWT proposes to run a cement bond log to evaluate the cement seal after each of the three casings are cemented. Before the well can be confirmed for use, WWT must also demonstrate to OGMD that the well and disposal formations are suitable for disposal and that the confining zone is adequate to confine disposed fluids.

Throughout the life of the well, WWT will be required to perform mechanical integrity tests. The requirement for internal mechanical integrity testing is annual, for external mechanical integrity testing it is every 5 years. If a well does not pass a mechanical integrity test, it may indicate the removable tubing or the packer is compromised. These components are the first line of defense (in addition to several others described above) in preventing leakage outside of the well that could impact fresh water or other resources. In such a case, replacing the tubing or the packer will remedy the problem and ensure the integrity of the well. OGMD has the authority to require additional mechanical integrity testing if there is work done on the well or if there is reason to believe there is a loss of mechanical integrity. If a well loses mechanical integrity, injection must cease until the operator can address the issue and demonstrate mechanical integrity.

WWT is not allowed to dispose of the leachate fluids at pressures that would fracture the formation. A test to determine disposal zone fracture pressure must be run before the well can be confirmed for use. A maximum injection pressure will be assigned which cannot be exceeded during disposal operations. WWT is required to continuously monitor the daily maximum and average injection pressures, rates, and volumes, pressures in the annulus between the innermost casing and the tubing, and any changes to the annulus pressure system. This information must be submitted to OGMD monthly.

Some commenters expressed concern regarding the identification and evaluation of an existing old abandoned well bore in the area. Part 625 rules require an evaluation within an Area of Review (AOR) extending one-quarter mile to two miles from the proposed injection well. This is done to identify any wells that penetrate the injection zone and may allow upward fluid migration based on how the wells were constructed and/or plugged. For this application OGMD identified only one existing, abandoned wellbore penetrating the proposed injection zone within two miles of the proposed well. This well was located greater than a mile from the proposed WWT 1 well. OGMD reviewed well construction and plugging records in its files for this well. The well was drilled in the 1940s and plugged to the standards existing at that time. WWT demonstrated in their application that their proposed injection pressures at the WWT 1 well will be low enough to prevent fluid migration into the freshwater zone at this old plugged wellbore, over a mile away. If migration of fluid through the confining zone is discovered, the Part 625 Mineral Well rules require the operator to immediately cease injection and submit a remedial plan to OGMD.

ALTERNATIVES

OGMD received comments regarding the current disposal of the landfill leachate and alternate treatment technologies. Some of the leachate currently goes to the City of Cadillac for wastewater treatment. Leachate that the landfill is unable to send for the wastewater treatment is instead sent to an existing Class I injection well for disposal. Many commenters shared their opinion that the leachate should continue to be treated at the wastewater treatment plant. The OGMD believes the proposed injection well provides better protection for the environment than sending the leachate to a wastewater treatment plant or trucking it to a remote disposal well.

OGMD DECISION

OGMD staff has conducted a thorough and intensive review of the proposed WWT Part 625 permit application and has determined that the application meets the applicable state regulatory requirements and that a permit should be issued. OGMD did not receive any comments identifying matters of fact that would dictate denial of a permit under Part 625 or its administrative rules. We appreciate all the comments that were received and the passion of those who have taken an interest in this proposal. The OGMD's final decision must follow established legal principles and be based upon whether or not an application adheres to statutes and rules. We recognize this decision may not satisfy all parties; however, we are confident that the decision is the correct one. OGMD believes that this document adequately responds to the concerns of those who submitted comments. The OGMD staff will monitor and oversee this project as it moves forward, and we will continue to hold the operator accountable and ensure WWT adheres to Part 625 regulations and requirements.